Prompt optimisation

OPRO

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GSM8K: grade-school maths word problems

Q: Alannah, Beatrix, and Queen are preparing for the new school year and have been given books by their parents. Alannah has 20 more books than Beatrix. Queen has 1/5 times more books than Alannah. If Beatrix has 30 books, how many books do the three have together?

Yang et al., Figure 3 and section 5

GSM8K: grade-school maths word problems

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The answer on file

140

Yang et al., Figure 3 and section 5

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A: The instruction goes here Then the working, then a number

At the start of the answer

The answer on file

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Right if they match

Yang et al., Figure 3 and section 5

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At the start of the answer

The answer on file

140

Right if they match



7,473 training problems

To optimise on (a 3.5% sample)



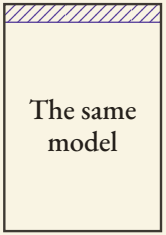
1,319 test problems

To check the result on

Yang et al., Figure 3 and section 5

Same model, different instruction

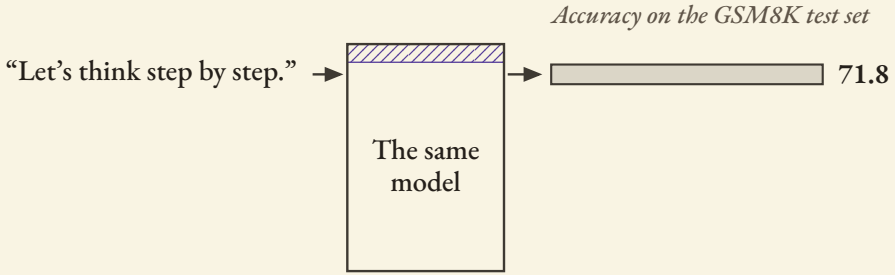
Accuracy on the GSM8K test set



PaLM 2-L

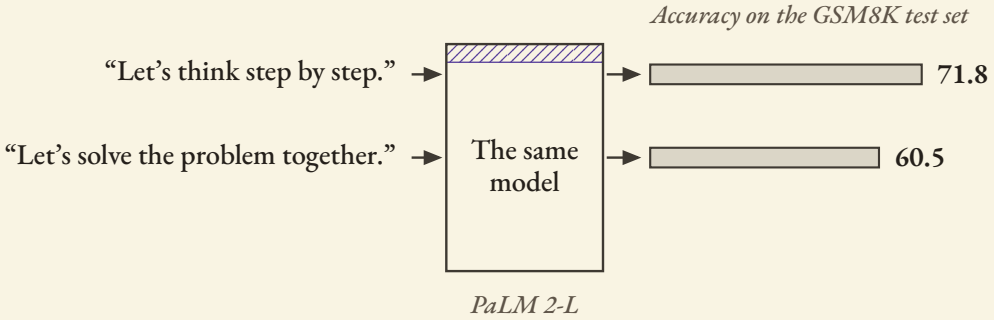
Yang et al., Large Language Models as Optimizers, section 5.2.3

Same model, different instruction



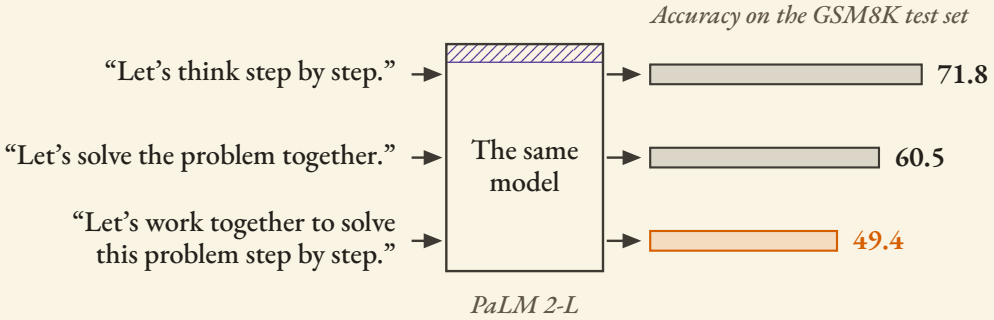
Yang et al., Large Language Models as Optimizers, section 5.2.3

Same model, different instruction



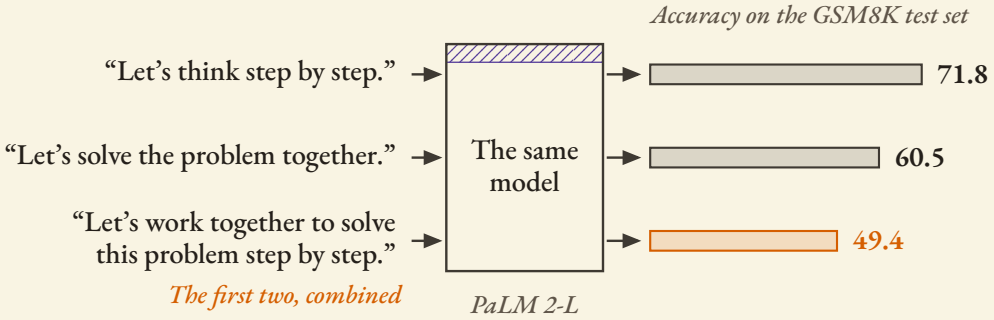
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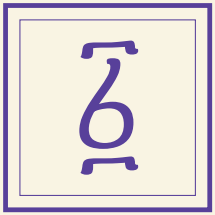


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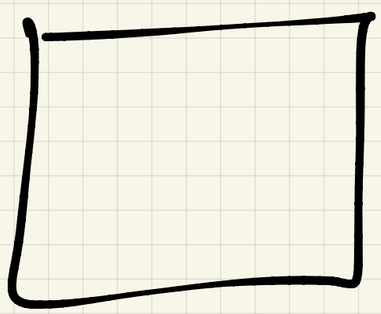
Yang et al., Large Language Models as Optimizers, section 5.2.3



Section 1

OPRO

OPRO



Pool of initial prompts

"please"

"think hard"

Accuracy

One step of OPRO

Meta-prompt

Past instructions,
with their scores
The best 20, worst first

Yang et al., sections 5.1 and 5.2.1

One step of OPRO

Meta-prompt

Past instructions,
with their scores
The best 20, worst first

Three example
problems, with answers
Picked at random

Yang et al., sections 5.1 and 5.2.1

One step of OPRO

Meta-prompt

Past instructions, with their scores
The best 20, worst first

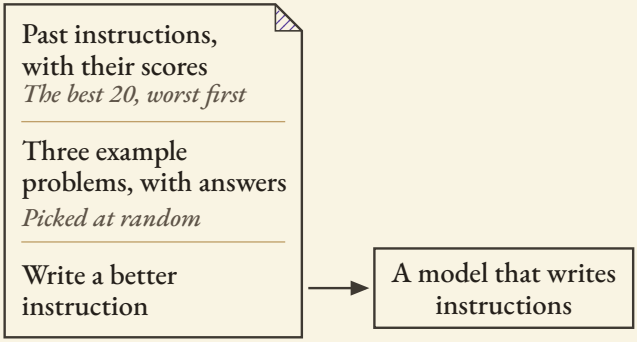
Three example problems, with answers
Picked at random

Write a better instruction

Yang et al., sections 5.1 and 5.2.1

One step of OPRO

Meta-prompt: what the writer reads



Optimiser: PaLM 2-L-IT

Yang et al., sections 5.1 and 5.2.1

One step of OPRO

Meta-prompt: what the writer reads

Past instructions,
with their scores
The best 20, worst first

Three example
problems, with answers
Picked at random

Write a better
instruction

A model that writes
instructions

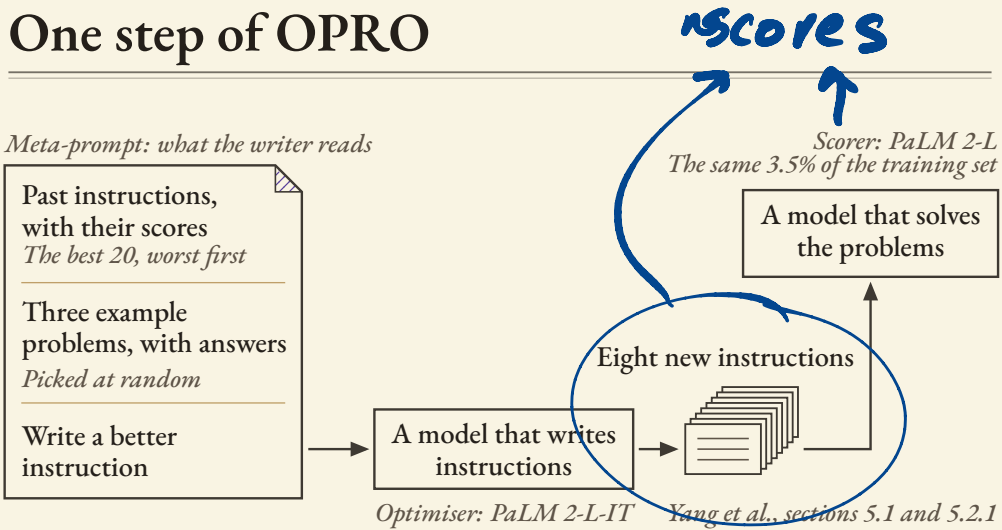
Eight new instructions



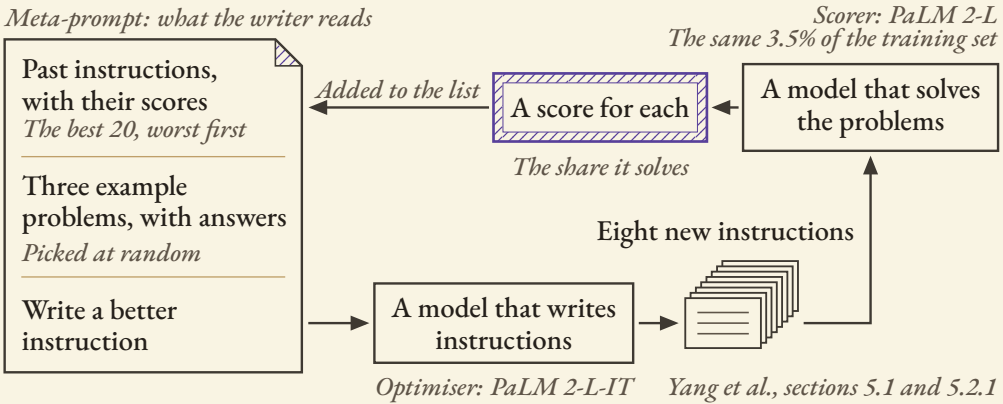
Optimiser: PaLM 2-L-IT

Yang et al., sections 5.1 and 5.2.1

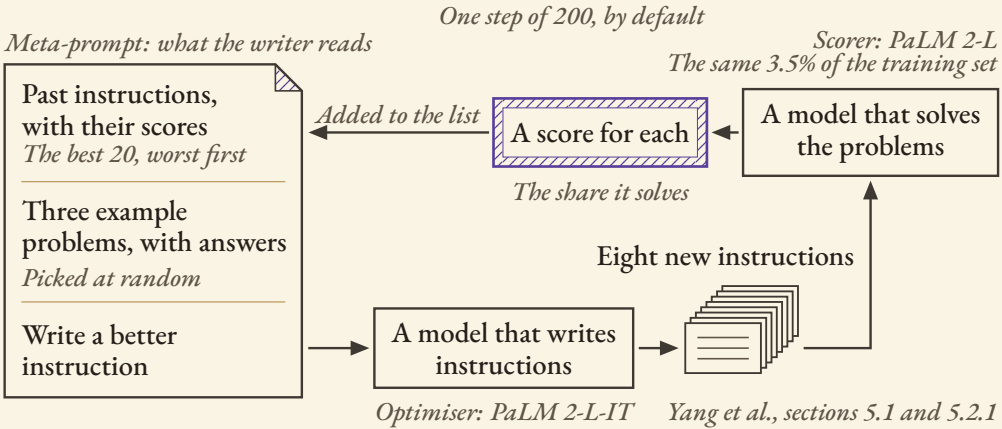
One step of OPRO



One step of OPRO



One step of OPRO



What the meta-prompt looks like

Yang et al., Figure 3

[Some texts and their scores, lowest score first]
text: Let's figure it out!
score: 61
text: Let's solve the problem.
score: 63
(. . . more instructions and scores . . .)

The instructions so far, worst first

What the meta-prompt looks like

Yang et al., Figure 3

[Some texts and their scores, lowest score first]
text: Let's figure it out!
score: 61
text: Let's solve the problem.
score: 63
(. . . more instructions and scores . . .)
[How a text is used: it replaces <INS>]
input: Q: Alannah, Beatrix, and Queen [. . .] how many books do the three have together?
A: <INS>
output: 140
(. . . more exemplars . . .)

The instructions so far, worst first

Example problems, with the slot marked

What the meta-prompt looks like

Yang et al., Figure 3

[Some texts and their scores, lowest score first]
text: Let's figure it out!
score: 61
text: Let's solve the problem.
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(. . . more instructions and scores . . .)
[How a text is used: it replaces <INS>]
input: Q: Alannah, Beatrix, and Queen [. . .] how many books do the three have together?
A: <INS>
output: 140
(. . . more exemplars . . .)
[Write a new text, unlike the old ones, with a score as high as possible]

The instructions so far, worst first

Example problems, with the slot marked

The request

What the meta-prompt looks like

Yang et al., Figure 3

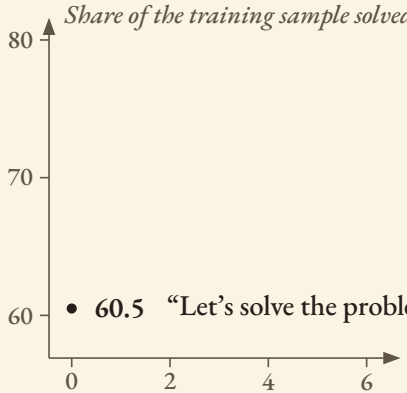
[Some texts and their scores, lowest score first]
text: Let's figure it out!
score: 61
text: Let's solve the problem.
score: 63
(. . . more instructions and scores . . .)
[How a text is used: it replaces <INS>]
input: Q: Alannah, Beatrix, and Queen [. . .] how many books do the three have together?
A: <INS>
output: 140
(. . . more exemplars . . .)
[Write a new text, unlike the old ones, with a score as high as possible]

The instructions so far, worst first
Worst first beat best first and random order

Example problems, with the slot marked

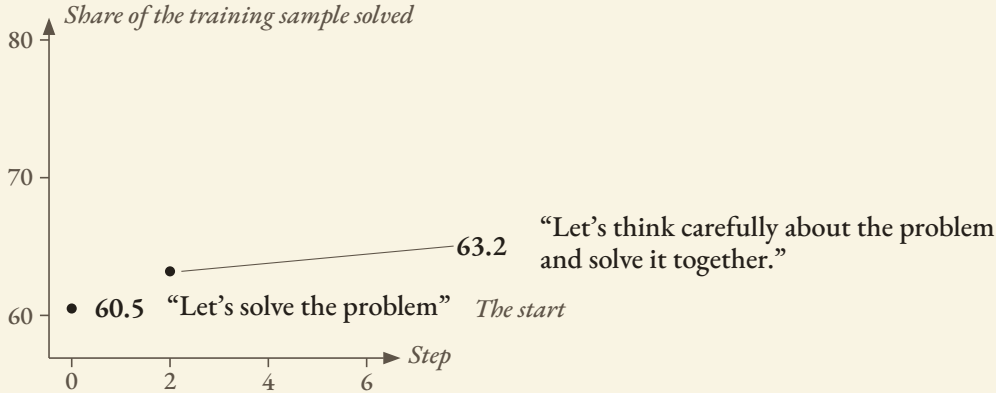
The request

Better instructions within six steps



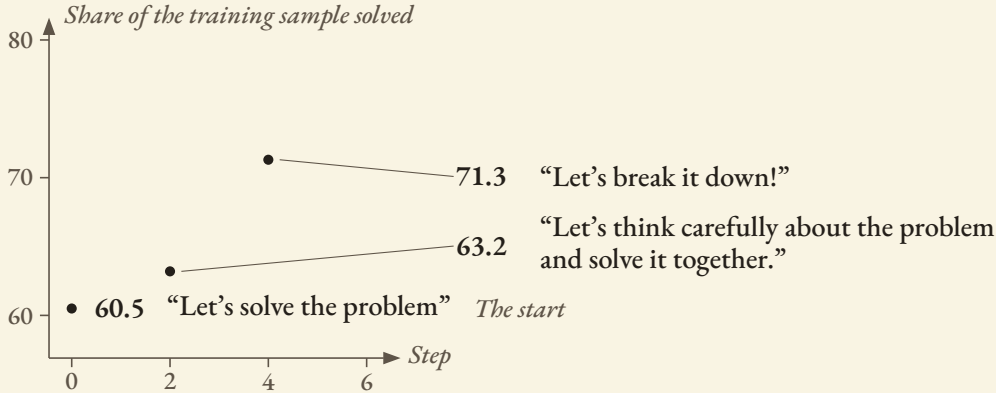
Yang et al., section 5.2.1, PaLM 2-L solving, PaLM 2-L-IT writing

Better instructions within six steps



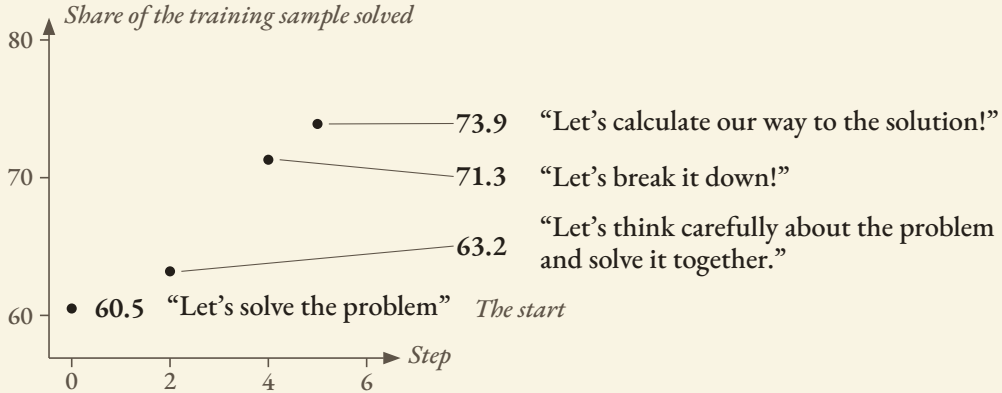
Yang et al., section 5.2.1, PaLM 2-L solving, PaLM 2-L-IT writing

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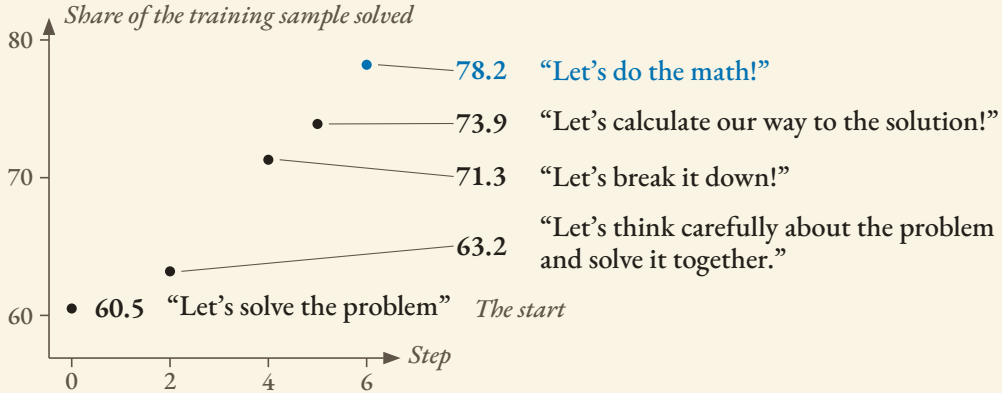
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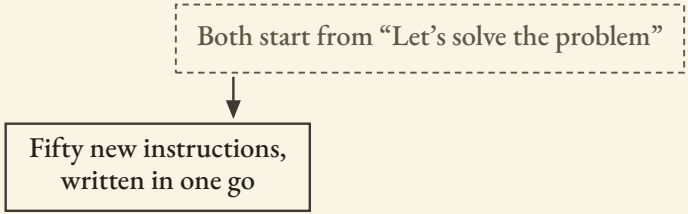
Yang et al., section 5.2.1, PaLM 2-L solving, PaLM 2-L-IT writing

Fifty at once against eight a step

Both start from “Let’s solve the problem”

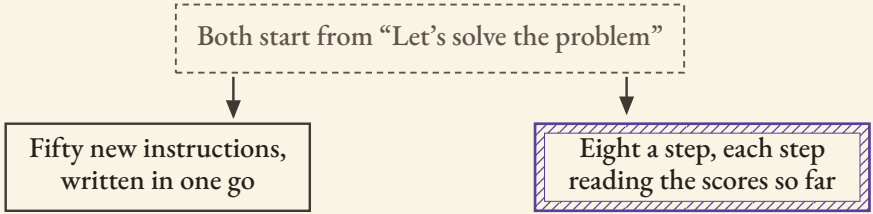
Yang et al., section 5.3

Fifty at once against eight a step



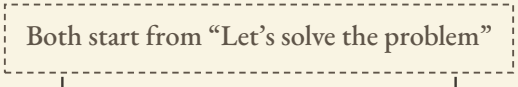
Yang et al., section 5.3

Fifty at once against eight a step



Yang et al., section 5.3

Fifty at once against eight a step



Fifty new instructions, written in one go

It never sees their scores

Eight a step, each step reading the scores so far

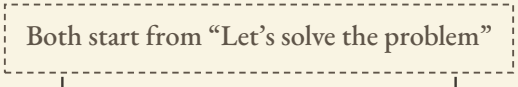
OPRO

Accuracy on the test set

60.8 Fifty at once: “Let’s solve the problem”, the start itself

Yang et al., section 5.3

Fifty at once against eight a step



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It never sees their scores

Eight a step, each step reading the scores so far

OPRO

Accuracy on the test set

60.8 Fifty at once: “Let’s solve the problem”, the start itself

76.3 OPRO, within six steps: “Let’s do the math!”

Yang et al., section 5.3

Eight instructions on GSM8K, ranked

Share of the GSM8K test set solved by PaLM 2-L, the instruction at the start of its answer

③			
④			
①			
⓪			
②			
⑥			
⑦			
⑧	**34.0**	“ ” *No instruction at all*	*Baseline*

Eight instructions on GSM8K, ranked

Share of the GSM8K test set solved by PaLM 2-L, the instruction at the start of its answer

⑤		
④		
③		
②		
①		
⑥		
⑦	58.8	"Let's work this out in a step by step way to be sure we have the right answer." — *Baseline, Zhou et al.*
⑧	34.0	" " *No instruction at all* — *Baseline*

Eight instructions on GSM8K, ranked

Share of the GSM8K test set solved by PaLM 2-L, the instruction at the start of its answer

3			
4			
5			
6			
7			
6	60.8	“Let’s solve the problem.”	*Baseline*
7	58.8	“Let’s work this out in a step by step way to be sure we have the right answer.”	*Baseline, Zhou et al.*
8	34.0	“ ” *No instruction at all*	*Baseline*

Eight instructions on GSM8K, ranked

Share of the GSM8K test set solved by PaLM 2-L, the instruction at the start of its answer

③			
④			
⑤			
⓪			
①	71.8	“Let’s think step by step.”	*Baseline, Kojima et al.*
⑥	60.8	“Let’s solve the problem.”	*Baseline*
⑦	58.8	“Let’s work this out in a step by step way to be sure we have the right answer.”	*Baseline, Zhou et al.*
⑧	34.0	“ ” *No instruction at all*	*Baseline*

Eight instructions on GSM8K, ranked

Share of the GSM8K test set solved by PaLM 2-L, the instruction at the start of its answer

③			
④			
⑤			
①	74.5	“Let’s combine our numerical command and clear thinking to quickly and accurately decipher the answer.”	*Found by gpt-4*
②	71.8	“Let’s think step by step.”	*Baseline, Kojima et al.*
⑥	60.8	“Let’s solve the problem.”	*Baseline*
⑦	58.8	“Let’s work this out in a step by step way to be sure we have the right answer.”	*Baseline, Zhou et al.*
⑧	34.0	“ ” *No instruction at all*	*Baseline*

Eight instructions on GSM8K, ranked

Share of the GSM8K test set solved by PaLM 2-L, the instruction at the start of its answer

3			
4			
5	78.5	"A little bit of arithmetic and a logical approach will help us quickly arrive at the solution to this problem."	*Found by gpt-3.5-turbo*
0	74.5	"Let's combine our numerical command and clear thinking to quickly and accurately decipher the answer."	*Found by gpt-4*
1	71.8	"Let's think step by step."	*Baseline, Kojima et al.*
6	60.8	"Let's solve the problem."	*Baseline*
7	58.8	"Let's work this out in a step by step way to be sure we have the right answer."	*Baseline, Zhou et al.*
8	34.0	" " *No instruction at all*	*Baseline*

Eight instructions on GSM8K, ranked

Share of the GSM8K test set solved by PaLM 2-L, the instruction at the start of its answer

③			
④	79.9	"Break this down."	*Found by PaLM 2-L*
⑤	78.5	"A little bit of arithmetic and a logical approach will help us quickly arrive at the solution to this problem."	*Found by gpt-3.5-turbo*
①	74.5	"Let's combine our numerical command and clear thinking to quickly and accurately decipher the answer."	*Found by gpt-4*
②	71.8	"Let's think step by step."	*Baseline, Kojima et al.*
⑥	60.8	"Let's solve the problem."	*Baseline*
⑦	58.8	"Let's work this out in a step by step way to be sure we have the right answer."	*Baseline, Zhou et al.*
⑧	34.0	" " *No instruction at all*	*Baseline*

Eight instructions on GSM8K, ranked

Share of the GSM8K test set solved by PaLM 2-L, the instruction at the start of its answer

🏆	80.2	**"Take a deep breath and work on this problem step-by-step."**	*Found by PaLM 2-L-IT*
⑨	79.9	"Break this down."	*Found by PaLM 2-L*
⑩	78.5	"A little bit of arithmetic and a logical approach will help us quickly arrive at the solution to this problem."	*Found by gpt-3.5-turbo*
⓪	74.5	"Let's combine our numerical command and clear thinking to quickly and accurately decipher the answer."	*Found by gpt-4*
⑧	71.8	"Let's think step by step."	*Baseline, Kojima et al.*
⑥	60.8	"Let's solve the problem."	*Baseline*
⑦	58.8	"Let's work this out in a step by step way to be sure we have the right answer."	*Baseline, Zhou et al.*
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